USB-C

USB1
TYPE-C 16PIN 2MD(073)

D1
USBLC6-2SC6

C20
10uF

R5
5.1kΩ

R6
5.1kΩ

ESD Protection

PWR LED

R8
470Ω

PWR
A-SP192DGHC-C01-4T

3.3V LDO

U4
AMS1117-3.3_C347222

C17
22uF

C16
10uF

1.8V LDO

U5
AMS1117-1.8_C5250991

C18
10uF

C19
10uF

NeoPixels

LED9
WS2812B-2020

LED10
WS2812B-2020

LED3
WS2812B-2020

LED1
WS2812B-2020

LED2
WS2812B-2020

LED4
WS2812B-2020

LED6
WS2812B-2020

LED5
WS2812B-2020

LED7
WS2812B-2020

LED8
WS2812B-2020

LED12
WS2812B-2020

LED13
WS2812B-2020

LED14
WS2812B-2020

HR Sensor

U2
MAX30102EFD+T_C6454833

R3
4.7kΩ

R4
4.7kΩ

R2
4.7kΩ

C14
100nF

C15
10uF

C12
4.7uF

C13
100nF

ESP32-C3

U1
ESP32-C3FH4

L3
2.2nH

L4
2.2nH

L1
24nH

C9
10uF

C8
100nF

C7
1uF

C10
3.3pF

C11
3.3pF

C1
10nF

C2
1uF

C3
10pF

C4
10pF

C5
100nF

C6
100nF

R7
5.1kΩ

R1
499Ω

U3
40MHz

ANT1
WIFI_ANTENNA

Sheet: /	
File: CARDio.kicad_sch	
Title: TITLE: ECG Business Card	
Size: User	Date:
KiCad E.D.A. 10.0.3	Rev: Id: 1/1
REV: 1.0	
Company: Ashrafy Industries	Sheet: 1/1
Date: 2025-07-29 Drawn By: Ali Ashrafy	